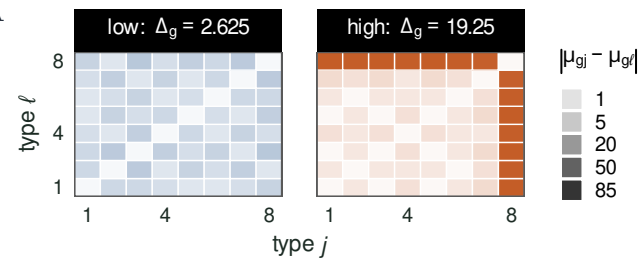
Gene g fixed, $J = 8$ cell types j, ℓ, type means (counts) μ_{gj}; both genes have $\bar{\mu}_g = 12.5$ and every $\mu_{gj} \geq 1$.
low-Gini gene $\mu_{g\cdot} = (12, 15, 9, 14, 11, 13, 10, 16)$ high-Gini gene $\mu_{g\cdot} = (1, 2, 1, 3, 1, 2, 4, 86)$

A Pairwise inequality

$$\Delta_g = \frac{1}{J^2} \sum_{j=1}^{J} \sum_{\ell=1}^{J} |\mu_{gj} - \mu_{g\ell}|$$

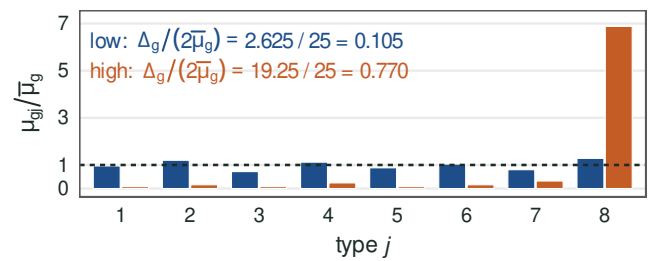
Mean absolute difference of the type means μ_{gj}; in counts, so it doubles when every μ_{gj} doubles.



B Scale-free

$$\mathrm{Gini}_g = \frac{\Delta_g}{2\bar{\mu}_g}, \qquad \bar{\mu}_g = \frac{1}{J} \sum_{j=1}^{J} \mu_{gj}$$

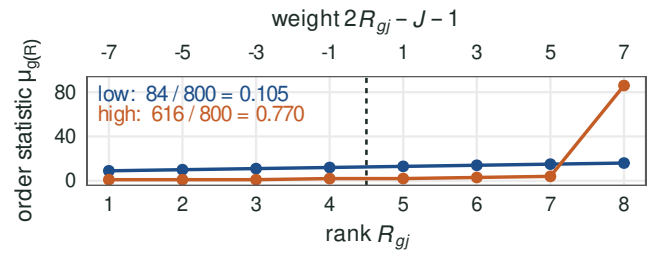
Dividing by twice the mean leaves Gini_g unchanged when all means are multiplied by a constant $c > 0$.



C Ordering

$$\mathrm{Gini}_g = \frac{1}{J^2 \bar{\mu}_g} \sum_{j=1}^{J} (2R_{gj} - J - 1)\, \mu_{gj}$$

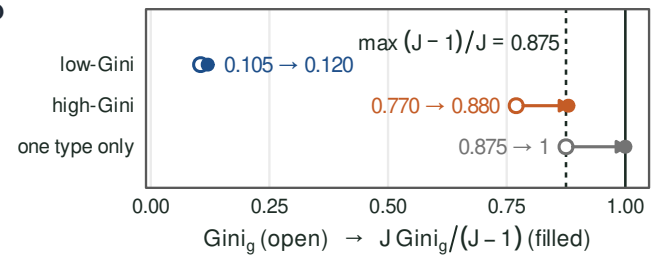
R_{gj}: rank of μ_{gj}; order statistics $\mu_{g(1)} \leq \cdots \leq \mu_{g(J)}$. One weighted sum over J sorted means replaces the J^2 pairs. Panel: the sum over $J^2 \bar{\mu}_g = 800$.



D Finite-J correction

$$\max \mathrm{Gini}_g = \frac{J-1}{J}, \qquad \frac{J}{J-1}\,\mathrm{Gini}_g \in [0, 1]$$

Reached when one type carries all the mass (grey row); the rescaled score of that gene is 1.



E Shares $u_{gj} = \mu_{gj} / \sum_\ell \mu_{g\ell}$

$$\mathrm{Gini}_g = \frac{1}{2J} \sum_{j=1}^{J} \sum_{\ell=1}^{J} |u_{gj} - u_{g\ell}|$$

$$= \frac{1}{J} \sum_{j=1}^{J} (2R_{gj} - J - 1)\, u_{gj}$$

Since $u_{gj} = \mu_{gj}/(J\bar{\mu}_g)$, B and C give this value.

